广東工業大學
GUANGDONG UNIVERSITY OF TECHNOLOGY

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# Resource Scheduling Strategies for Cooperative Perception and Computing in the Internet of Vehicles

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## Introduction

### Challenges

- **Limited resources**: Vehicles and Roadside Units (RSUs) have constrained computing and caching capacities, making efficient task scheduling challenging.
- **Dynamic network**: Rapid topology changes and heterogeneous hardware complicate real-time offloading and coordination.
- **Multi-stage cooperation**: Cooperative perception involves raw, feature, and object-level data, requiring coordinated scheduling across multiple stages..
- **Latency-sensitive feedback**: Task results must be returned to real-time control and decision-making.

### Objective

- To develop efficient resource scheduling strategies for cooperative perception and computing in the Internet of Vehicles (IoV), minimizing latency and optimizing edge resource utilization.

### Contributions

- **Vehicle–road–cloud collaborative framework** for joint service caching and computation offloading.
- **Multi-stage cooperative perception framework** for task offloading and resource allocation.
- **Low-complexity deep reinforcement learning (DRL) method** for near-optimal, adaptive scheduling.
- **Late-fusion-based prototype implementation** validating feasibility and real-world performance.

## Problem Formulation

**P1**: Design a joint service caching and computation offloading scheme to minimize the long-term average task latency in vehicle–road–cloud systems.

$$\min_{\{c_{v,k}^{V}(t),c_{e,k}^{E}(t),o_{v,k}^{V}(t),o_{e,k}^{E}(t)\}} \sum_{t=1}^{t_{end}} \gamma^{t-1}\left(\frac{T_d(t)}{T_{dmax}}\right)$$

$$\begin{aligned}
&s.t.\\
&c_{v,k}^{V}(t)\in\{0,1\},\forall v\in V,\forall k\in K,\forall t\in\{1,2,\ldots,t_{end}\};\\
&c_{e,k}^{E}(t)\in\{0,1\},\forall e\in E,k,t;\\
&0\le o_{v,k}^{V}(t)\le 1,\forall v,k,t;\\
&0\le o_{e,k}^{E}(t)\le 1,\forall e,k,t;\\
&0\le\gamma\le 1;\\
&\textstyle\sum_{k=1}^{N_K} c_{v,k}^{V}(t)\theta_k\le S_v^V,\forall v,t;\\
&\textstyle\sum_{k=1}^{N_K} c_{e,k}^{E}(t)\theta_k\le S_e^E,\forall e,t.
\end{aligned}$$

**P2**: Design a cooperative task offloading and resource allocation scheme to minimize the delay of region-of-interest (RoI) perception tasks under latency and resource constraints.

$$\min_{\{x_i,o_{ij},\mu_{ij}\}} D_{ego}^{RoI}$$

$$\begin{aligned}
&s.\ t.\\
&D_{v_i}(x_i,o_{ij},\mu_{ij})\le min\{\tau_i^{tole},\tau_{j,ego}^{hold}\},\\
&\textstyle\sum_{j=1}^{|\Omega_{server}^{RoI}|} o_{i,j}=1, v_i\in\Omega_v^{RoI}, s_j\in\Omega_{server}^{RoI},\\
&\textstyle\sum_{i=1}^{|\Omega_v^{RoI}|}\mu_{ij}\le 1, v_i\in\Omega_v^{RoI}, s_j\in\Omega_{server}^{RoI},\\
&x_i\in\{0,1,2\}, o_{ij}\in\{0,1\},\\
&\mu_{ij}\in[0,1].
\end{aligned}$$

## Late-Fusion-Based Prototype Implementation

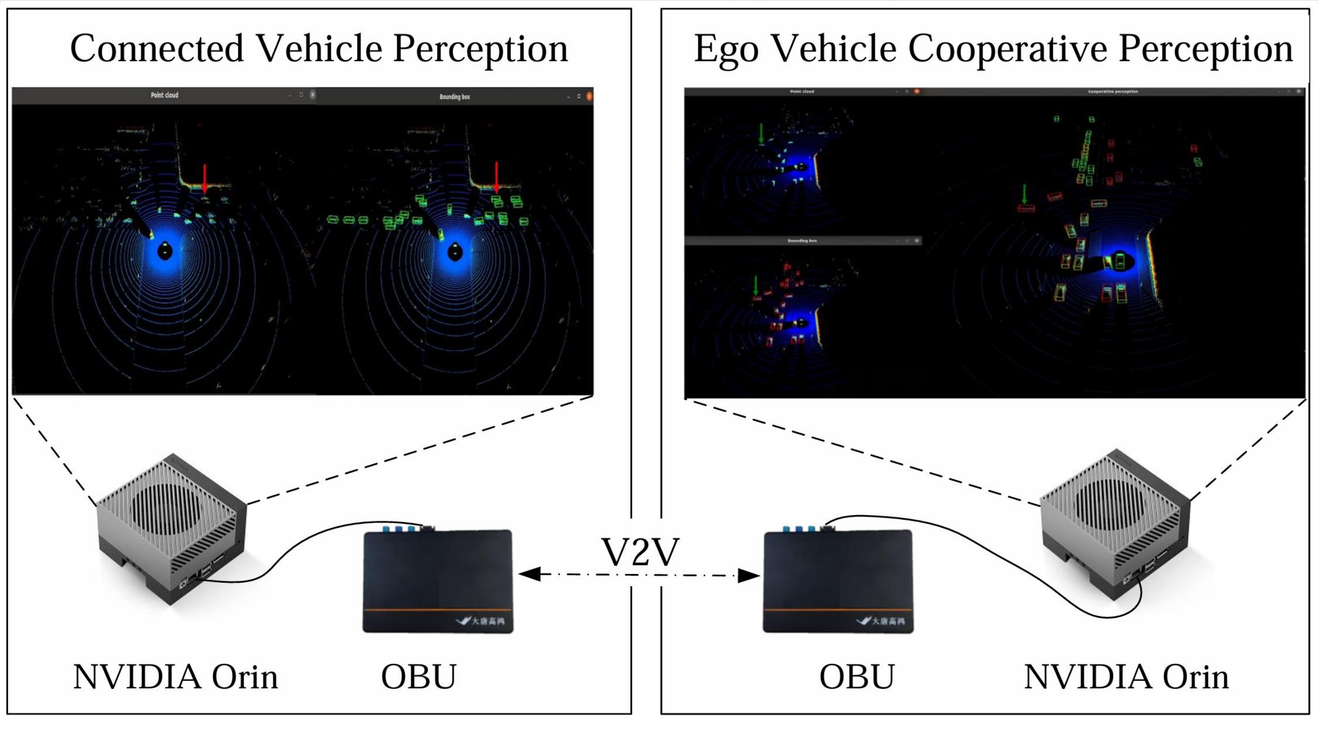
Connected Vehicle Perception
Ego Vehicle Cooperative Perception
V2V
NVIDIA Orin
OBU
OBU
NVIDIA Orin

Fig. 5. Late-fusion cooperative perception system developed based on the NVIDIA Orin platform and on-board units (OBUs).

## Cooperative Perception

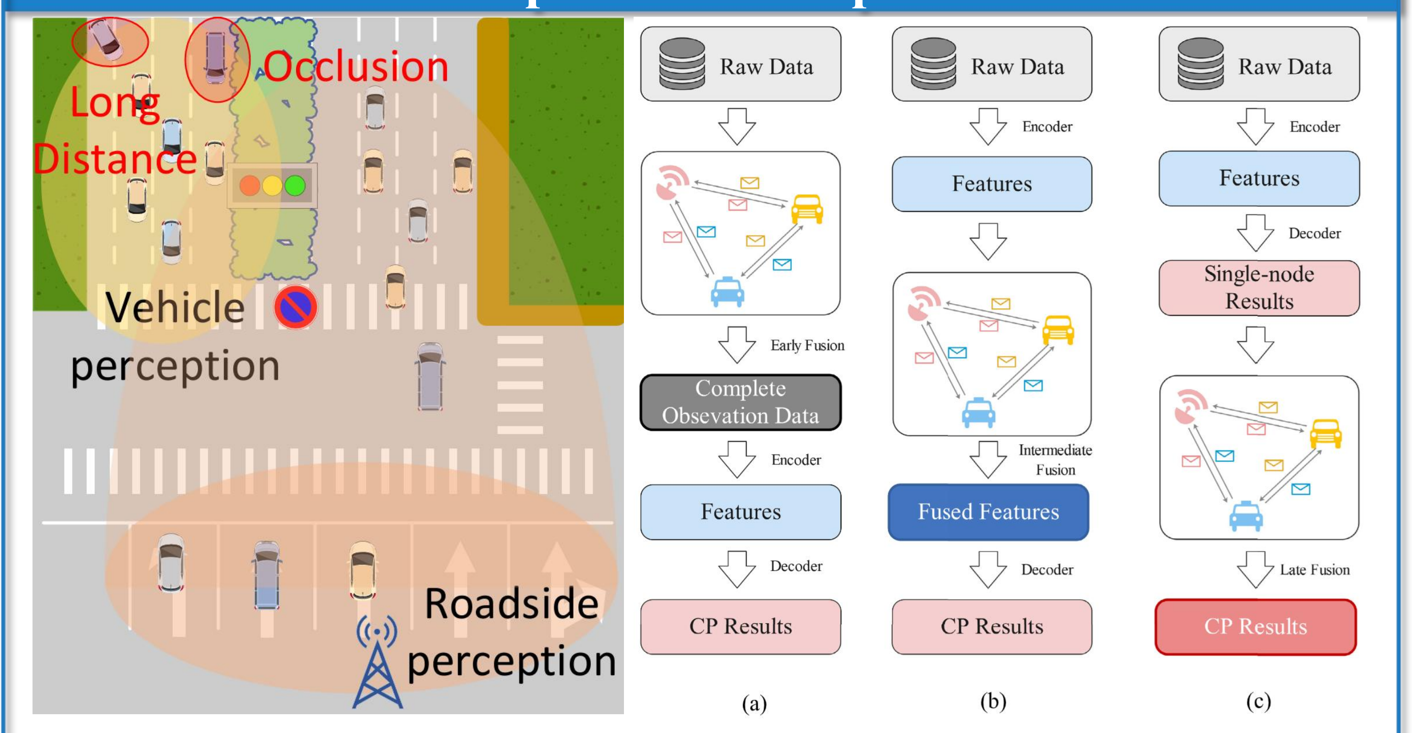

Fig. 1. Illustration of multi-view cooperative perception and collaboration strategies: (a) early fusion; (b) intermediate fusion; and (c) late fusion

## Cooperative Perception and Computation Model

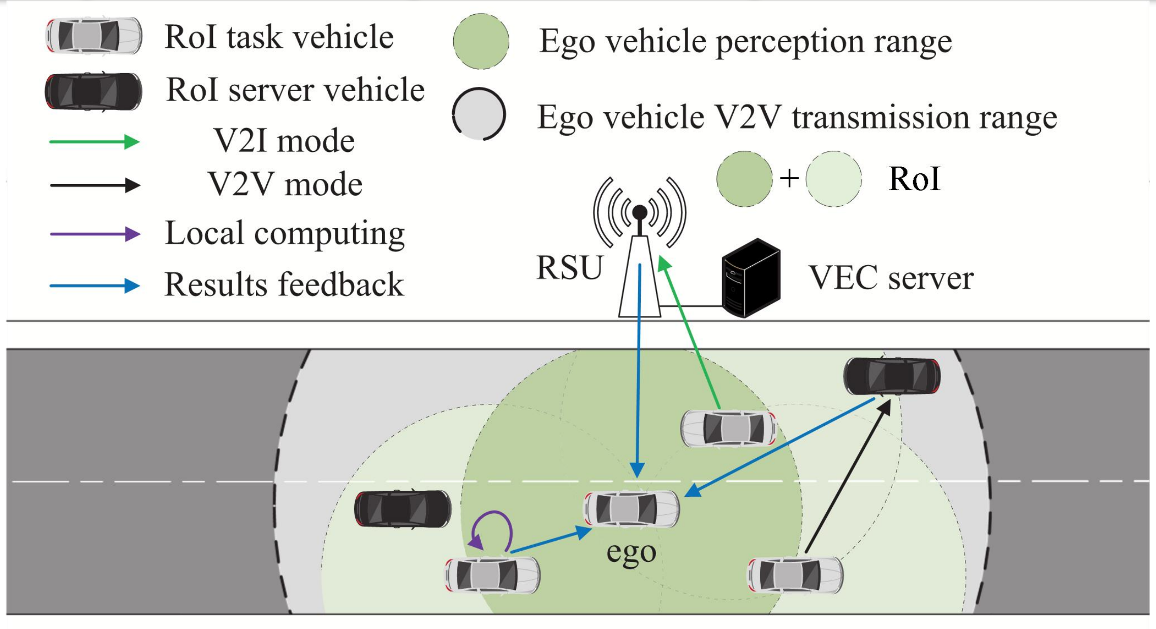

Fig. 2. An illustration of the cooperative perception and computation model in vehicular edge computing (VEC).

## Proposed Methods

The **DDPG-based edge caching and offloading scheme** iteratively observes the network state, selects caching and offloading actions via the actor network, receives rewards based on task completion delay and cache hit ratio, and updates the actor-critic networks to improve future decisions. This allows the system to adaptively learn efficient caching and offloading strategies in dynamic vehicular networks.

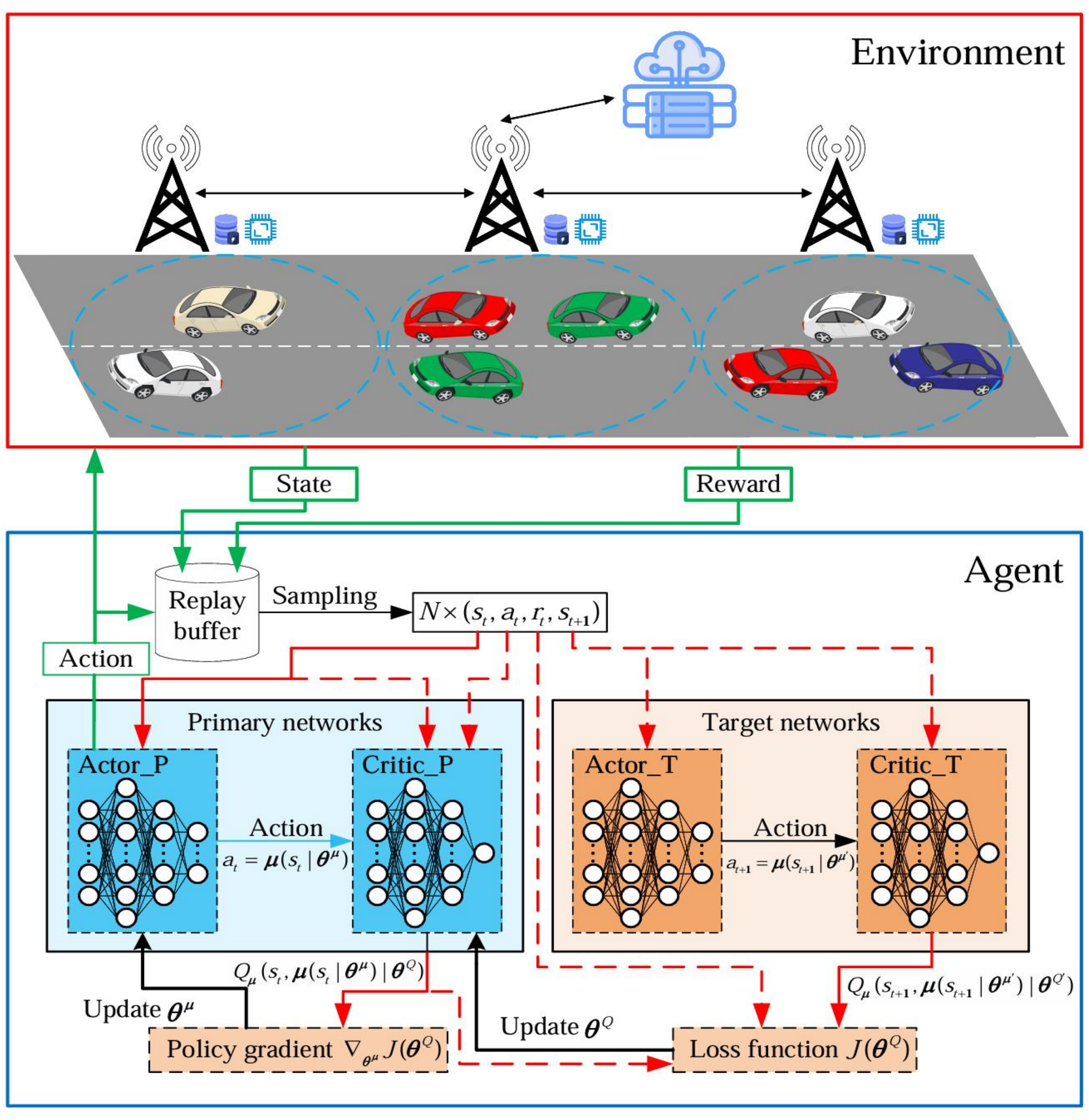
Environment
State
Reward
Agent
Replay buffer
Sampling
$N\times(s_t,a_t,r_t,s_{t+1})$
Action
Primary networks
Actor_P
Critic_P
Action
$a_t=\mu(s_t\mid\theta^{\mu})$
Target networks
Actor_T
Critic_T
Action
$a_{t+1}=\mu(s_{t+1}\mid\theta^{\mu'})$
$Q_{\mu}(s_t,\mu(s_t\mid\theta^{\mu})\mid\theta^{Q})$
$Q_{\mu}(s_{t+1},\mu(s_{t+1}\mid\theta^{\mu'})\mid\theta^{Q'})$
Update $\theta^{\mu}$
Update $\theta^{Q}$
Policy gradient $\nabla_{\theta^{\mu}}J(\theta^{Q})$
Loss function $J(\theta^{Q})$

Fig. 3. DDPG-Based Edge Caching and Offloading Scheme

**The OTORA algorithm** exhaustively searches feasible offloading and caching decisions, verifies resource constraints, and then employs a nonlinear programming (NLP) solver to obtain the optimal task scheduling and resource allocation.

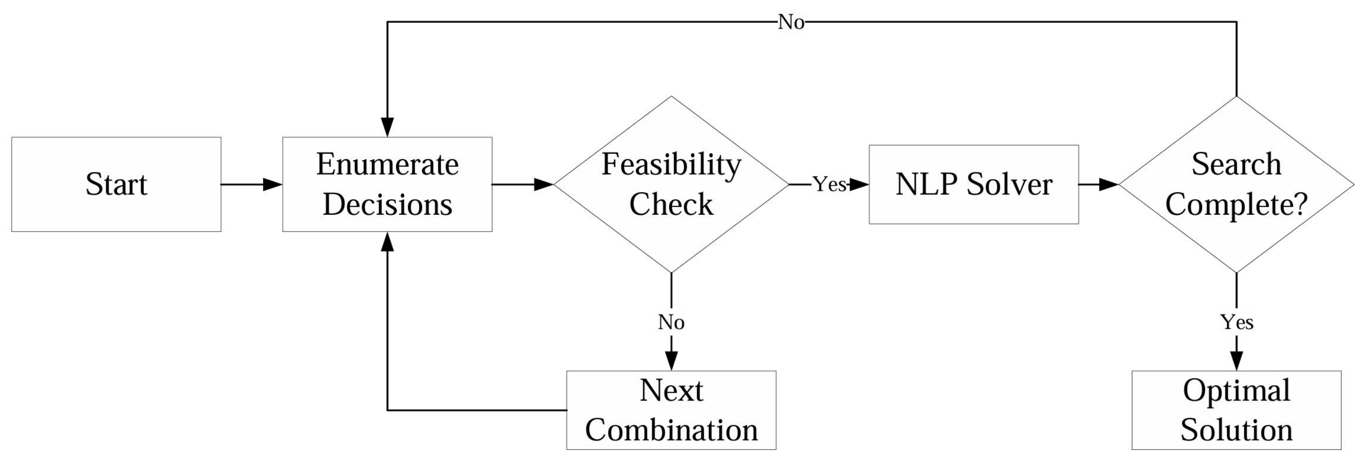

Fig. 4. Optimal task offloading and resource allocation (OTORA) algorithm

## Simulation Results

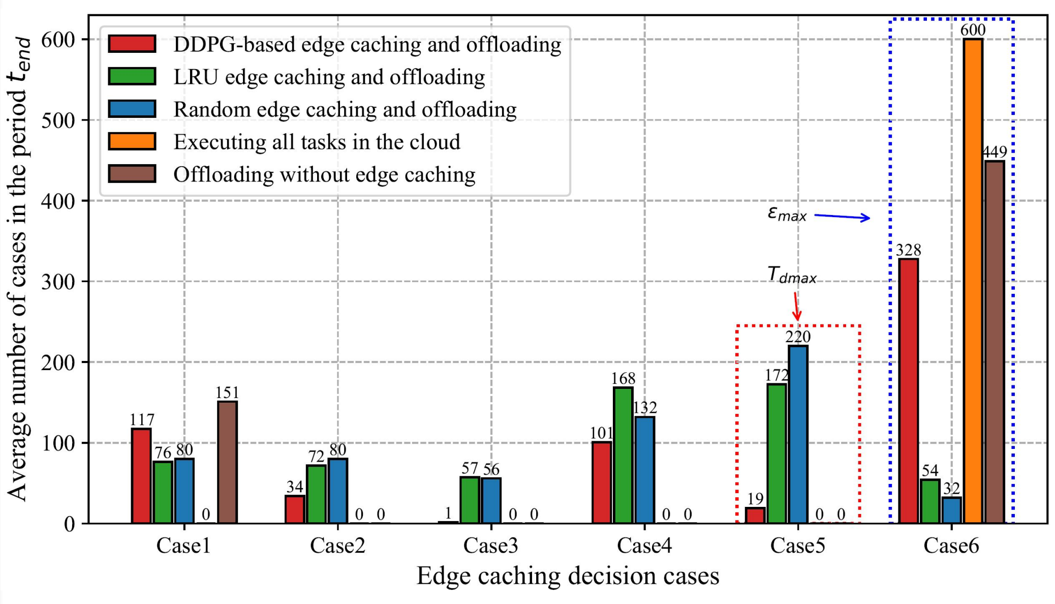

Fig. 6. Comparison of caching decision distributions under different schemes.

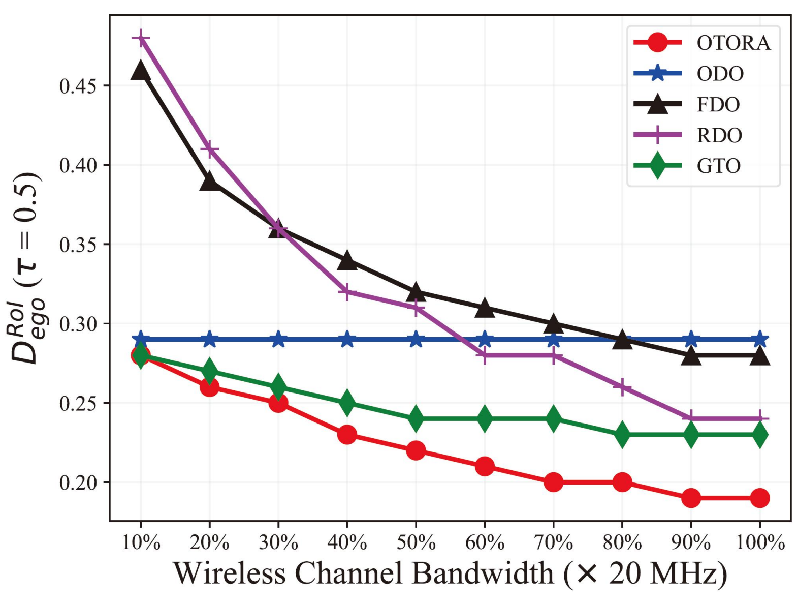

Fig. 7. Impact of wireless channel bandwidth on RoI task delay under different offloading schemes.

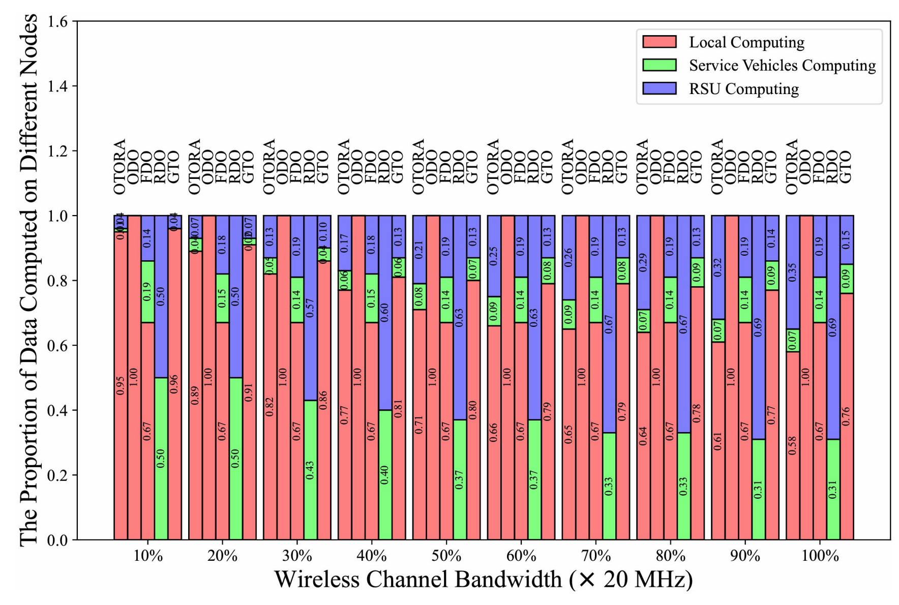

Fig. 8. Effect of wireless channel bandwidth increase on data computation proportions among various nodes.

## References

[1] Z. Xue, C. Liu, F. Wen and G. Han, "Joint Optimization of Task Offloading and Resource Allocation for Cooperative Perception in Vehicular Edge Computing Systems," in *IEEE Transactions on Vehicular Technology*, early access, 2025, doi: 10.1109/TVT.2025.3586857.
[2] Z. Xue, C. Liu, C. Liao, G. Han and Z. Sheng, "Joint Service Caching and Computation Offloading Scheme Based on Deep Reinforcement Learning in Vehicular Edge Computing Systems," in *IEEE Transactions on Vehicular Technology*, vol. 72, no. 5, pp. 6709-6722, May 2023.
[3] Z. Song, T. Xie, H. Zhang, J. Liu, F. Wen and J. Li, "A Spatial Calibration Method for Robust Cooperative Perception," in *IEEE Robotics and Automation Letters*, vol. 9, no. 5, pp. 4011-4018, May 2024.